Solutions of Tutorial 6

Sol^n 1(i)

$$\lim_{x\to 0}\frac{\tan x}{x} = \lim_{x\to 0}\frac{\sin x}{x}\cdot\frac{1}{\cos x}$$

$$= 1 \qquad \left[\because \lim_{x\to 0}\frac{\sin x}{x}=1\right]$$

(ii) $$\lim_{x\to 0}\frac{\sqrt{1+x}-1}{x} = \lim_{x\to 0}\frac{\cancel{1}+x-\cancel{1}}{x(\sqrt{1+x}+1)}$$

$$= \lim_{x\to 0}\frac{1}{\sqrt{1+x}+1}$$

$$= \frac{1}{2}$$

(iii) $$\lim_{x\to 1}\frac{x^2-1}{x-1} = \lim_{x\to 1}\frac{(x+1)\cancel{(x-1)}}{\cancel{(x-1)}} = 2$$

(iv) $$\lim_{x\to a}\frac{\sin x-\sin a}{x-a} = \lim_{x\to a}\frac{2\cos\left(\frac{x+a}{2}\right)\cdot\sin\left(\frac{x-a}{2}\right)}{x-a}$$

$$= \lim_{y\to 0}\frac{2\cos\left(\frac{y+2a}{2}\right)\sin\left(\frac{y}{2}\right)}{y} \qquad \left[\text{Take } x-a=y\right.$$

$$= \cos a$$

Sol^n 2(i) Let $f(x) = x+\operatorname{sgn}(x)$

$$L.H.L = \lim_{x\to 0^-} f(x) = \lim_{x\to 0^-}(x-1) = -1$$

$$R.H.L = \lim_{x\to 0^+} f(x) = \lim_{x\to 0^+}(x+1) = 1$$

$\because$ L.H.L $\neq$ R.H.L. So $\lim_{x\to 0} f(x)$ does not exist

(ii) Let $f(x) = e^{\frac{1}{x}}$

$$L.H.L = \lim_{h \to 0} f(0-h) = \lim_{h \to 0} e^{-\frac{1}{h}} = 0$$

$$R.H.L = \lim_{h \to 0} f(0+h) = \lim_{h \to 0} e^{\frac{1}{h}} = \infty$$

$\because$ $L.H.L \neq R.H.L$. So $\lim_{x \to 0} f(x)$ does not exist.

(iii) Let $f(x) = \frac{\sin x}{|x|}$

$$L.H.L = \lim_{h \to 0} f(0-h) = \lim_{h \to 0} \frac{\sin(-h)}{-(-h)} = -\lim_{h \to 0} \frac{\sin h}{h}$$
$$= -1$$

$$R.H.L = \lim_{h \to 0} f(0+h) = \lim_{h \to 0} \frac{\sin h}{h} = 1$$

$\because$ $L.H.L \neq R.H.L$. Thus $\lim_{x \to 0} f(x)$ does not exist.

v) $$f(x) = \begin{cases} x+3, & x \leq 2 \\ -2x+5, & x > 2 \end{cases}$$

$$L.H.L = \lim_{h \to 0} f(2-h) = \lim_{h \to 0} (2-h)+3 = 5$$

$$R.H.L = \lim_{h \to 0} f(2+h) = \lim_{h \to 0} -2(2+h)+5 = 1$$

$\because$ $L.H.L \neq R.H.L$. Thus $\lim_{x \to 2} f(x)$ does not exist.

Sol^n 3

$$f(x) = \begin{cases} -x^2, & x \le 0 \\ 5x-4, & 0 < x \le 1 \\ 4x^2-3x, & 1 < x \le 2 \\ 3x+4, & x \geq 2 \end{cases}$$

for continuity at $x=0$,

$$L.H.L = \lim_{h \to 0} f(0-h) = \lim_{h \to 0} -h^2 = 0$$

$$R.H.L = \lim_{h \to 0} f(0+h) = \lim_{h \to 0} 5h-4 = -4$$

$\because$ L.H.L $\neq$ R.H.L. So $\lim_{x \to 0} f(x)$ does not exist.

Thus function is not continuous at $x=0$ and jump discontinuity is there.

for continuity at $x=1$,

$$L.H.L = \lim_{h \to 0} 5(1-h)-4 = 1$$

$$R.H.L = \lim_{h \to 0} 4(1+h)^2 - 3(1+h) = 1$$

$\because$ L.H.L. = R.H.L = $1 = f(1)$

Thus fun is continuous at $x=1$.

for continuity at $x=2$.

$$LHL = \lim_{h \to 0} 4(2-h)^2 - 3(2-h) = 10$$

$$R.H.L = \lim_{h \to 0} 3(2+h)+4 = 10$$

$\because$ L.H.L = R.H.L = $10 = f(2)$

Thus fun is continuous at $x=2$.

Sol^n 4

$$f(x) = \begin{cases} a+bx^3, & x<1 \\ 4, & x=1 \\ b-ax^2, & x>1 \end{cases}$$

$$L.H.L = \lim_{h\to 0} f(1-h) = \lim_{h\to 0} a+b(1-h)^3 = a+b$$

$$R.H.L = \lim_{h\to 0} f(1+h) = \lim_{h\to 0} b-a(1+h)^2 = b-a$$

Given that fun^n is cont^n at $x=1$. Thus, we have

$L.H.L = R.H.L = f(1)$

$a+b = 4$ ——— ①

$b-a = 4$ ——— ②

On solving equ^n ① and equ^n ②, we get

$a = 0$, $b = 4$.

Sol^n 5(i)

$$f(x) = \begin{cases} x^2 \sin\frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$f'(0) = \lim_{h\to 0} \frac{f(h)-f(0)}{h}$$

$$= \lim_{h\to 0} \frac{h^2 \sin\frac{1}{h}}{h}$$

$$= \lim_{h\to 0} h \sin\frac{1}{h}$$

$$= 0$$

Thus, fun^n is diff-ble at $x=0$.

(ii) $f(x) = \begin{cases} e^{-\frac{1}{x^2}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

$$f'(0) = \lim_{h \to 0} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \to 0} \frac{e^{-\frac{1}{h^2}} - 0}{h}$$

$$= 0$$

(iii) $f(x) = e^{-|x|} = \begin{cases} e^{x}, & x \geq 0 \\ e^{-x}, & x < 0 \end{cases}$

$$Lf'(0) = \lim_{h \to 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \to 0} \frac{e^{h} - 1}{-h}$$

$$= -\lim_{h \to 0} \frac{e^{h} - 1}{h}$$

$$= -1$$

$$Rf'(0) = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \to 0} \frac{e^{h} - 1}{h}$$

$$= 1$$

$\therefore$ $Lf'(0) \neq Rf'(0)$. Thus fn is not diff'ble at $x = 0$. So $f'(0)$ does not exist.